

Problem K

Subwords

Time limit: 2 seconds

Problem Description

You have a grammar with the start symbol S , a set of variables V , a set of terminal symbols T , and a set of rules P . Every upper-case letter is a variable, so $V = \{A, \dots, Z\}$, and every lower-case letter is a terminal symbol, so $T = \{a, \dots, z\}$. A rule is of the form $A \rightarrow \alpha$, where $A \in V$ is a variable and α is a string consisting of variables and terminal symbols. The grammar never contains chain rules of the form $A \rightarrow B$, where the right side consists of a single variable. Rules of the form $A \rightarrow b$ are possible, where the right side consists of a single terminal.

A word can be derived by starting with $\alpha = S$ and then repeatedly choosing some variable X in α , some rule $X \rightarrow \beta$, and then replacing one X in α with β . To be precise, if $\alpha = \alpha'X\alpha''$ and we have a rule $X \rightarrow \beta$, then we can replace α with $\alpha'\beta\alpha''$, and we write $\alpha \Rightarrow \alpha'\beta\alpha''$.

Here is a complete example. The grammar has the rules

$$S \rightarrow SA, \quad S \rightarrow b, \quad A \rightarrow bS, \quad A \rightarrow a.$$

Then we can derive the word bbb in four steps:

$$S \Rightarrow SA \Rightarrow bA \Rightarrow bbS \Rightarrow bbb.$$

On the other hand, the word aaa cannot be derived.

Your task is to answer the following question for a grammar G and a word w (consisting of terminal symbols):

- Can you add zero or more terminal symbols at the beginning of w and zero or more terminal symbols at the end of w such that the resulting word uwv can be derived by G ?

Input Format

The first line contains an integer k , representing the number of rules.

Followed by k lines, each of which contains one rule. A rule is encoded by its left side, followed by $\rightarrow$, and then followed by its right side.

The last line contains the word w , representing the input word of the question.

Output Format

Output a single line containing Yes or No that answers the question.

Technical Specification

- $1 \leq k \leq 25$
- The left side of every rule is an upper-case letter.
- The right side of every rule is a string of upper-case and lower-case letters. Moreover, it is either a string of length at least 2 or a single lower-case letter.
- The total length of the right sides of all rules is at most 100.
- The length of w is at least 1 and at most 400, and w consists of lower-case letters.

Sample Input 1

```
4
S -> SA
S -> b
A -> bS
A -> a
bbb
```

Sample Output 1

```
Yes
```

Sample Input 2

```
2
S -> aSb
S -> ab
ba
```

Sample Output 2

```
No
```

Sample Input 3

```
2
S -> aSa
S -> b
ab
```

Sample Output 3

```
Yes
```